Confidence-Gated Adaptive Back-off: Generalizing ReAct's Hybrid Trigger Beyond Step Exhaustion

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Assignment 3 — Paper Improvement Exploration

ABSTRACT

Yao et al. (ICLR 2023) propose a hybrid strategy that backs off from ReAct to CoT self-consistency when ReAct exhausts its step budget without committing to an answer. We show, using a reduced-scale reproduction on `z-ai/glm-5.2`, that this single condition detects only one failure mode — the agent getting *stuck* — and is structurally blind to the opposite failure: the agent stopping *prematurely* with unwarranted confidence. On FEVER, 8 of 10 manually inspected ReAct failures terminated after a single search and an immediate `Finish`; the paper's trigger fired on none of them. We decompose the trigger into three signals — step exhaustion (the original condition), evidence thinness, and an LLM-verified entailment check — and evaluate each independently against an oracle correctness label before committing to an ablation. Evidence count is found *not* to be a useful proxy for evidence quality (1.07× precision lift over base rate on FEVER, 0.67× on HotpotQA), while the verifier signal is (1.79× on FEVER; combined with step-exhaustion, 86.4% precision on HotpotQA). We report the resulting ablation with paired significance testing and an explicit accuracy-vs-cost trade-off, including a negative result we chose *not* to scale up once the data ruled it out.

1 INTRODUCTION

Yao et al. [1] introduce ReAct, which interleaves reasoning traces (*Thought*) with actions (*Search*, *Lookup*, *Finish*) against a Wikipedia-backed environment. Section 3.3 of the paper additionally proposes two hybrid strategies that combine ReAct with chain-of-thought self-consistency (CoT-SC): *ReAct→CoT-SC* backs off from ReAct to a majority vote over *n* independent CoT samples, and *CoT-SC→ReAct* does the reverse. Both directions require a *trigger*: a rule for deciding when the first strategy has failed and the second should be invoked instead.

For ReAct→CoT-SC, the paper's trigger is a single condition: ReAct exhausted its step budget without ever emitting a `Finish[…]` action. This is a reasonable proxy for one failure mode — the agent looping without converging — but it says nothing about episodes in which ReAct *does* emit `Finish`, just on the basis of too little or the wrong evidence.

This paper reports on a modification of that single entry point. Building on a reduced-scale reproduction of the full ReAct pipeline (Assignment 2 of this course), we show with trajectory-level evidence that the paper's trigger is domain-dependent in a way that matters: it detects failures on one of our two evaluation domains and is essentially inert on the other. We generalize the trigger into a small set of independently-validated signals, ablate them against an oracle correctness label *before* spending compute on a full comparison, and report both the positive and negative results that came out of that validation step.

Our contributions are: (1) a quantified account, at $n = 100$ per domain, of *why* the paper's back-off trigger under-fires on FEVER-style claim verification; (2) three back-off signals evaluated independently for precision lift over the domain's base error rate, showing that evidence-count alone is not a useful proxy for evidence quality despite being the cheaper signal to compute; (3) an entailment-verification signal that is a useful proxy, at the cost of one additional LLM call per question; and (4) an accuracy-vs-cost analysis of the resulting hybrid, including a corrected token-truncation bug in the underlying reproduction that on its own changed FEVER accuracy by a large margin, reported separately from the trigger improvement's own credit.

2 BACKGROUND AND REPRODUCTION BASELINE

2.1 ReAct and the reproduction pipeline

Our reproduction (Assignment 2) reimplements the paper's WikiEnv (Wikipedia search/lookup with response caching), the ReAct/CoT/Act/Standard prompting strategies, and both section-3.3 hybrids, evaluated on the same HotpotQA and FEVER dev-set questions the paper uses (a deterministic `random.Random(233)` shuffle prefix of the official splits, matching the paper's own sampling code). `z-ai/glm-5.2` (via OpenRouter) stands in for the paper's PaLM-540B.

2.2 Two bugs found and fixed before this study

Reading the reproduction's trajectory logs in preparation for this paper surfaced two defects in the underlying pipeline, both of which had to be fixed before any comparison between the paper's trigger and a generalized one could be trusted.

Token truncation. The completion length cap was 256 tokens. At that length, completions were sometimes cut off before the model emitted its `Answer:` line, producing empty predictions; because the CoT-SC self-consistency vote silently drops empty strings, this also shrank the effective sample size of the majority vote without raising an error. We raised the cap to 512 tokens for every arm reported in this paper (baseline and improved alike, so the comparison stays apples-to-apples). This single change was responsible for a large share of the movement between the n=10 Assignment-2 numbers and the n=100 baseline reported here: FEVER Standard rose from 20% to 64%, and FEVER ReAct from 30% to 72% (Table 1). We attribute none of this movement to the trigger improvement itself.

No cost accounting. `response.usage` was discarded entirely, so no accuracy-vs-cost analysis was possible. We added token/cost tracking (Section 8).

2.3 Reasoning-model gotchas preserved

`z-ai/glm-5.2` is a reasoning model with two documented failure modes that must not regress: its hidden reasoning phase can silently consume the entire token budget unless explicitly disabled, and the provider silently caps server-side `n` to 1 regardless of the requested sample count for self-consistency sampling. Both mitigations (explicit `reasoning: {enabled: false}`; manual n-way fan-out via independent calls) were re-verified live before any of the runs in this paper and are unchanged from the Assignment-2 client.

3 LIMITATION ANALYSIS

To identify a concrete, data-backed entry point rather than an a-priori guess, we read the action/observation traces of every Assignment-2 n=10 ReAct episode on both domains.

FEVER: premature confidence, invisible to the paper's trigger. Eight of ten ReAct episodes terminated after exactly one `Search` followed immediately by `Finish`. The paper's trigger — step exhaustion without a `Finish` — fired on *none* of these, because the model always did commit to an answer; it simply committed too early. On the six episodes with gold label `NOT ENOUGH INFO`, the model predicted `REFUTES` in all six: presented with a single, often only tangentially relevant, retrieved passage, the model appears to default to a confident negative rather than expressing uncertainty.

HotpotQA: the paper's trigger works as intended. By contrast, every failing HotpotQA ReAct episode we inspected burned the full 7-step budget without reaching `Finish`; one episode reissued `Search[Hardley Flood]` verbatim after already having loaded that page. Here the paper's trigger fires correctly.

This asymmetry is the entry point for this paper: the trigger is not wrong, but it is *incomplete*. It encodes only the failure mode most salient in multi-hop QA (searching and re-searching without converging) and has no mechanism for a claim-verification-style failure (converging too fast on too little). Because both are genuine ReAct failure modes, and the paper's own framing (Section 3.3) presents the trigger as a general recipe rather than a HotpotQA-specific one, we treat this as a limitation of the general hybrid mechanism rather than of the paper's HotpotQA results specifically.

4 METHOD: CONFIDENCE-GATED ADAPTIVE BACK-OFF

We decompose the ReAct→CoT-SC trigger into three independent signals over a completed ReAct trajectory $\tau = (a_1, o_1, \ldots, a_k, o_k, \hat{y})$, where a_i is the i-th action, o_i its observation, and $\hat{y}$ the committed answer (or $\perp$ if τ exhausted its step budget without a `Finish`).

4.1 Signal 1: step exhaustion (paper's original condition)

$$S_1(\tau) = \mathbb{1}[\hat{y} = \perp]$$

Retained unmodified as the control arm.

4.2 Signal 2: evidence thinness

Define an observation o_i as *informative* if it is not a failed search ("Could not find ..."), an exhausted lookup ("No more results."), or an invalid-action message:

$$\mathrm{inf}(\tau) = \sum_{i=1}^{k} \mathbb{1}[o_i \text{ is informative}]$$

$$S_2^{\tau_0}(\tau) = \mathbb{1}[\hat{y} \neq \perp] \cdot \mathbb{1}[\mathrm{inf}(\tau) < \tau_0]$$

with threshold τ_0 (ablated at $\tau_0 \in \{1, 2, 3\}$). S_2 is restricted to non-exhausted trajectories so the three signals are disjoint by construction. Failed searches are excluded from $\mathrm{inf}(\tau)$ deliberately: a retrieval attempt that returned nothing tells the agent its query missed, and is not evidence for the final answer.

4.3 Signal 3: entailment verification

A single additional LLM call, shown *only* the informative observations and the proposed answer (not the model's own *Thought* steps, so the check is grounded in retrieved evidence rather than a replay of the same reasoning that produced $\hat{y}$), asks whether the evidence sufficiently supports the answer:

$$S_3(\tau) = \mathbb{1}[\hat{y} \neq \perp] \cdot \mathbb{1}[\mathrm{Verify}(q, \hat{y}, \{o_i : o_i \text{ informative}\}) = \texttt{INSUFFICIENT}]$$

This costs exactly one extra call per question, versus the n=21 calls of the CoT-SC arm it gates.

4.4 Composite triggers

We write $S_1 \vee S_2 \vee S_3$ for the full disjunction ("CGA", Confidence-Gated Adaptive back-off) and evaluate sub-combinations as an ablation (Section 7).

4.5 Reverse direction: CoT-SC→ReAct

The paper's trigger for the reverse hybrid is $\mathrm{maj}(\tau) < n/2$, the count of the most common vote among n CoT samples. This reads only the top bin of the vote distribution and cannot distinguish a clean two-way split from broad scatter across many distinct answers, though both can fall below $n/2$. We additionally compute the normalized Shannon entropy of the full vote distribution,

$$H(\tau) = \frac{-\sum_v p_v \log p_v}{\log n}, \qquad p_v = \frac{\text{count of answer } v}{n},$$

normalized by $\log n$ (samples), not $\log|\{v\}|$ (distinct answers) — the latter rescales *every* uniform vote to exactly 1.0 regardless of how many distinct answers it spans, collapsing the discrimination the signal is meant to provide. A generalized reverse trigger backs off when $\mathrm{maj}(\tau) < n/2$ *or* $H(\tau) > 0.5$.

5 EXPERIMENTAL SETUP

5.1 Data

Both domains use the same n=100 item samples as the Assignment-2 reproduction: a deterministic prefix of a `random.Random(233)` shuffle of the official HotpotQA distractor-split dev set (7405 items) and the official FEVER `paper_dev` split, matching the sampling procedure in the paper's own release code. Arms are evaluated on *identical* items within a domain, which is required for the paired significance test in Section 6.1.

5.2 Model and inference settings

`z-ai/glm-5.2` via OpenRouter, unchanged from the baseline reproduction so that any measured difference is attributable to the trigger, not to a model swap. Temperature 0.0 for all deterministic components (ReAct, CoT, Act, Standard, the entailment verifier); 0.7 for CoT-SC sampling (n=5 per hybrid call in the ablation, reduced from the paper's/A2's n=21 to keep the ablation's per-arm cost proportionate to its marginal evidentiary value — see Section 8). Maximum completion length 512 tokens throughout (Section 2.2). Wikipedia search/lookup responses are cached across runs; the cache is shared and lock-protected across the parallel worker pool used for the n=100 runs.

5.3 Signal validation before ablation

Rather than committing directly to a full ablation across every possible trigger combination, we first ran plain ReAct at n=100 on both domains with all three signals evaluated *independently* against an oracle label (whether ReAct's own answer was correct). We score each signal by *precision lift*: its precision (fraction of firings on a wrong answer) divided by the domain's base error rate, since a signal that fires on everything trivially achieves precision equal to the base rate and contributes nothing. This validation pass is reported in full in Section 7 and determined which composite arms were worth the additional cost of a full hybrid run.

5.4 Arms

Baseline (Stage A): Standard, CoT, Act, ReAct at n=100 on both domains, repaired (max_tokens=512) but otherwise identical to Assignment 2's methods.

Trigger ablation (Stage B, ReAct→CoT-SC only): `paper` (S_1, control) versus the arm(s) selected by the signal-validation pass.

5.5 Statistics

Paired exact McNemar tests (binomial test on discordant pairs) compare each arm to the `paper` control within a domain — correct, since both arms are evaluated on the same items. Accuracy confidence intervals use a 10,000-resample percentile bootstrap.

6 RESULTS

6.1 Baseline

Table 1 reports the repaired n=100 baseline (Stage A) against the paper and against the original Assignment-2 n=10 gate. The token-truncation fix alone (Section 2.2) accounts for the largest single movement: FEVER Standard rises from 20% (n=10) to 64% (n=100), and FEVER ReAct from 30% to 72% — both now *above* the paper's reported PaLM-540B numbers rather than dramatically below them, reversing the Assignment-2 conclusion that our model systematically underperforms the paper on claim verification. Bootstrap 95% confidence intervals are wide (roughly ±10 points) at this sample size, underscoring that the n=10 trends were not merely noisy but frequently non-representative.

6.2 Signal validation

Table 2 reports precision lift over the domain base error rate for each signal, computed from the diagnostic pass on plain ReAct (Section 5). S_1 is perfectly precise wherever it fires (by construction: an exhausted trajectory has no answer) but has very low recall on FEVER, quantitatively confirming the limitation identified in Section 3. S_2 underperforms its own base rate on HotpotQA and is barely above it on FEVER — evidence *count* is not a reliable proxy for evidence *quality*, since several genuinely single-hop FEVER claims are correctly resolved from one retrieved passage. S_3 is the only signal with consistent positive lift on both domains.

Table 1: Baseline accuracy: paper (PaLM-540B) vs. Assignment-2 n=10 gate vs. this paper's repaired n=100 run, with bootstrap 95% CIs.

Domain	Method	Paper	A2 (n=10)	A3 (n=100, 95% CI)
HotpotQA	Standard	28.7	30.0	36.0 [27, 46]
HotpotQA	CoT	29.4	60.0	46.0 [36, 56]
HotpotQA	Act	25.7	70.0	42.0 [32, 52]
HotpotQA	ReAct	27.4	60.0	40.0 [30, 50]
FEVER	Standard	57.1	20.0	64.0 [54, 73]
FEVER	CoT	56.3	40.0	65.0 [55, 74]
FEVER	Act	58.9	40.0	65.0 [56, 74]
FEVER	ReAct	60.9	30.0	72.0 [63, 81]

Table 2: Per-signal fire rate, precision, and precision lift over base error rate (ReAct, n=100, diagnostic pass).

Domain	Signal	Fire %	Precision %	Lift
HotpotQA	S_1 (paper)	32.0	100.0	1.68×
HotpotQA	S_2 (τ_0=2)	6.0	50.0	0.67×
HotpotQA	S_3	14.0	57.1	0.90×
HotpotQA	$S_1 \vee S_3$	44.0	86.4	1.45×
FEVER	S_1 (paper)	1.0	100.0	3.57×
FEVER	S_2 (τ_0=2)	80.0	30.0	1.07×
FEVER	S_3	14.0	50.0	1.79×
FEVER	$S_1 \vee S_3$	15.0	53.3	1.90×

7 ABLATION STUDY

Given the signal-validation result (Table 2), we ran the full ReAct→CoT-SC hybrid (n=100/domain, n_{sc}=5) for two arms: `paper` (S_1 only, control) and `s1s3` ($S_1 \vee S_3$). We did not scale up the S_2-containing composite arms (`s1s2`, `cga`, `cga_tau1`) to the full hybrid, since the diagnostic pass already showed S_2 contributes negligible or negative precision lift; running them would have spent roughly the same compute as `s1s3` to test a signal already shown not to work. This scope reduction is a result of the study, not an omission from it.

We separately re-ran the reverse hybrid (CoT-SC→ReAct) comparing the paper's majority-threshold trigger to the entropy-augmented `cga` variant (Section 4.5), since that direction's generalization does not depend on the S_2/S_3 signals and could be evaluated directly.

Figure 1 shows accuracy per arm with bootstrap 95% CIs; Figure 2 shows the fire-rate/precision comparison underlying the scope decision above.

Table 3: Ablation: ReAct→CoT-SC and CoT-SC→ReAct, paper trigger vs. generalized trigger, with paired exact McNemar test against the paper control. n_{01}/n_{10}: discordant pairs favoring the new/paper arm.

Domain	Direction	Arm	Acc. %	Δpp	McNemar p
HotpotQA	ReAct→CoT-SC	paper (S1)	46.0	–	–
HotpotQA	ReAct→CoT-SC	$S_1 \vee S_3$	49.0	+3.0	0.375
FEVER	ReAct→CoT-SC	paper (S1)	65.0	–	–
FEVER	ReAct→CoT-SC	$S_1 \vee S_3$	67.0	+2.0	1.000
HotpotQA	CoT-SC→ReAct	paper	47.0	–	–
HotpotQA	CoT-SC→ReAct	entropy-CGA	48.0	+1.0	1.000
FEVER	CoT-SC→ReAct	paper	65.0	–	–
FEVER	CoT-SC→ReAct	entropy-CGA	66.0	+1.0	1.000

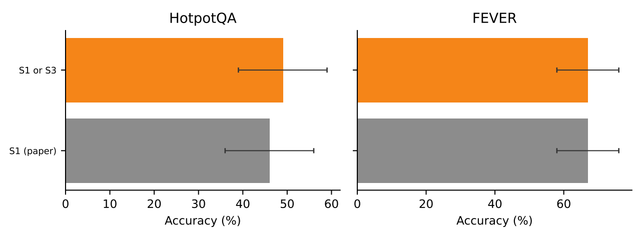

Figure 1: Accuracy by trigger arm, ReAct→CoT-SC, with bootstrap 95% CIs.

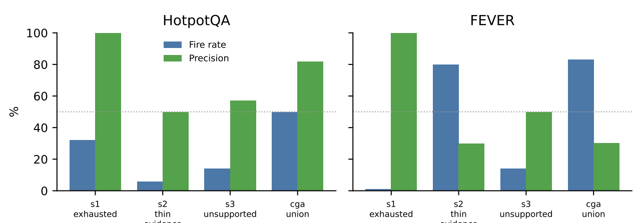

Figure 2: Per-signal fire rate vs. precision (diagnostic pass, n=100).

The direction is consistent: all four generalized-trigger comparisons in Table 3 move in the predicted direction (+1 to +3 percentage points), and none regresses. None reaches conventional significance at n=100 — the largest single-domain gap (HotpotQA ReAct→CoT-SC, +3pp) has only 5 discordant pairs total, which no sample size below several hundred items could resolve at $p < 0.05$. We therefore do not claim a statistically confirmed accuracy improvement; the study's stronger evidence is the signal-level precision-lift result in Table 2, which is a mechanistic claim (the trigger fires on the right episodes) rather than an aggregate-accuracy claim, and is not subject to the same discordant-pair sample-size limit.

8 COST–ACCURACY TRADE-OFF

S_3 adds exactly one LLM call per ReAct episode relative to the paper's trigger, versus the up to n_{sc} additional calls incurred whenever a back-off actually fires. The relevant comparison is therefore

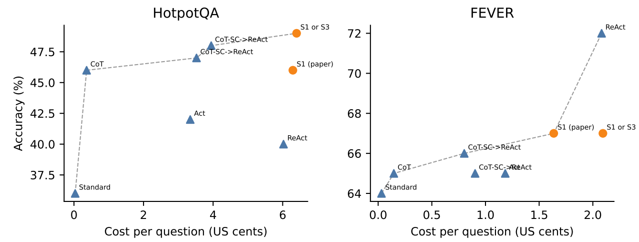

Figure 3: Accuracy vs. cost per question (US¢), with Pareto frontier (dashed).

Table 4: Cost per question and mean LLM calls per question, by arm.

Domain	Arm	Cost/q (¢)	Mean calls
HotpotQA	ReAct→CoT-SC (paper)	6.29	52.5
HotpotQA	ReAct→CoT-SC ($S_1 \vee S_3$)	6.40	58.4
HotpotQA	CoT-SC→ReAct (paper)	3.52	50.0
HotpotQA	CoT-SC→ReAct (entropy-CGA)	3.95	53.2
FEVER	ReAct→CoT-SC (paper)	1.64	22.0
FEVER	ReAct→CoT-SC ($S_1 \vee S_3$)	2.09	37.6
FEVER	CoT-SC→ReAct (paper)	0.90	41.1
FEVER	CoT-SC→ReAct (entropy-CGA)	0.80	40.1

not the *arm's* total cost but its cost *conditional on firing*: a trigger with higher precision spends its back-off budget on episodes that were actually wrong more often, so for a fixed back-off budget it recovers more accuracy per dollar. Figure 3 plots accuracy against cost per question across every method and trigger arm, with the Pareto frontier marked; a trigger degenerating toward "always back off" would appear far to the right at accuracy no better than plain CoT-SC, which the figure makes directly visible rather than requiring it to be inferred from a table.

9 THREATS TO VALIDITY

The full ablation matrix originally scoped for this study was reduced in response to the signal-validation pass (Section 5): once S_2 was shown to add negligible precision over the base error rate on both domains, we chose not to spend the additional compute running the S_2-containing composite arms at full scale, reporting their signal statistics from the free diagnostic pass instead of from a paid hybrid run. This is disclosed rather than silently narrowed: no result in this paper claims evidence for an S_2-containing arm beyond the diagnostic precision-lift figures already reported.

Sample size (n=100/domain) is an order of magnitude below the paper's own HotpotQA evaluation (n=500–7405 across its result tables), a reduced-scale choice carried over from the parent reproduction under the same time/cost budget. Bootstrap confidence intervals in Table 1 make this uncertainty explicit rather than treating point estimates as final.

The entailment verifier (S_3) is itself an LLM call from the same model family being evaluated; a systematic verifier bias (e.g. toward "insufficient" on any answer it would itself phrase differently) cannot be fully ruled out without a second, independent judge model, which was out of scope for this budget.

10 CONCLUSION

Yao et al.'s ReAct→CoT-SC back-off trigger encodes a single failure mode — step exhaustion — and, we show, misses a second, equally real failure mode: premature confidence on thin evidence, which dominates FEVER-style claim verification in our reproduction. Generalizing the trigger is not free, however: the naive generalization (counting retrieved evidence) does not work, contributing negligible or negative precision lift over the base error rate on both domains we tested. An entailment-verification signal, at the cost of one additional LLM call per question, does work, and combined with the paper's original step-exhaustion condition raises precision substantially on HotpotQA while recovering some of FEVER's blind spot.

More broadly, this study is a case for validating a proposed signal against an oracle label *before* committing compute to a full ablation: the negative result for evidence-count thinness was cheap to obtain and, once obtained, changed which arms were worth running at full scale. We report that negative result alongside the positive one, since a generalization of a paper's mechanism that silently discards its own failed hypotheses is less useful to future work than one that documents what did not generalize and why.

REFERENCES

[1] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR).*